Linear Algebraic Equations and Matrices (Ch 8)

Matrix notation: $[A]_{m \times n}$

a_{ij}: individual element of a matrix: i^{th} row, j^{th} column

Row vector: $[b] = [b_1 \; b_2 \cdots b_n]$

Column vector: $[c] = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$

Square matrix: $m = n$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

→ main diagonal (principal diagonal)

Symmetric matrix: $a_{ij} = a_{ji}$

Diagonal matrix: off-diagonal elements are zero

Identity matrix: a diagonal matrix with main diagonal elements equal to 1.

$$[A][I] = [I][A] = [A]$$

Upper triangular matrix:

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Lower triangular matrix

$$[A] = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Banded Matrix :

band

$$[A] = \begin{bmatrix} a_{11} & a_{12} & . & . \\ a_{21} & a_{22} & a_{23} & 0 \\ . & a_{32} & a_{33} & a_{34} \\ . & . & a_{43} & a_{44} \end{bmatrix}$$

↳ has bandwidth 3

tridiagonal matrix

Matrix Operating Rules

Equal $[A] = [B]$ iff $a_{ij} = b_{ij} \quad \forall\ i,j$

Addition $[C] = [A] \pm [B]$ $\quad c_{ij} = a_{ij} \pm b_{ij}$

Subtraction

Commutative : $[A] + [B] = [B] + [A]$

Associative : $([A] + [B]) + [C] = [A] + ([B] + [C])$

Multiplication $[D] = g[A]$, $d_{ij} = g a_{ij}$ (g: constant)

Product $[C] = [A][B]$ $\quad c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

$$[A]_{m \times n} \quad [B]_{n \times l} = [C]_{m \times l}$$

if inner dimensions are the same, matrix multiplication is possible

multiplication is **associative** : (if dimensions are suitable)

$$([A][B])[C] = [A]([B][C])$$

multiplication is **distributive** :

$$([A] + [B])[C] = [A][C] + [B][C]$$

not commutative:

$$[A][B] \neq [B][A]$$

(order of multiplication is important)

Inverse: if $[A]$ is square and nonsingular

$$[A][A]^{-1} = [A]^{-1}[A] = [I]$$

$[A]^{-1}$: inverse of $[A]$

2x2 matrix $\quad \bar{A}^{-1} = \dfrac{1}{a_{11}a_{22} - a_{21}a_{12}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Transpose: $A^T \qquad [A^T]_{ij} = a_{ji}$

Permutation Matrix (Transposition matrix): an identity matrix with rows and columns interchanged.

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$: switched row 1 and 3 of an identity matrix

PA: will switch the corresponding rows of A

$$PA = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -7 & 4 \\ 8 & 3 & -6 \\ 5 & 1 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 1 & 9 \\ 8 & 3 & -6 \\ 2 & -7 & 4 \end{bmatrix}$$

Augmentation: add columns

- useful when we perform identical operations on two matrix:

$$\left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array}\right]$$

Representing Linear Algebraic Eq^{n}s in matrix form

3x3 set of linear equations:

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}\right\} \quad Ax = b$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

To solve for x

$$A^{-1}Ax = A^{-1}b$$

($A^{-1}A = I$)

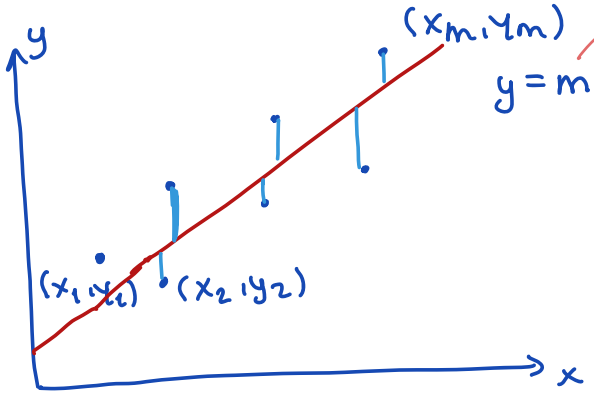
$$x = A^{-1}b$$

- Not a very efficient way (if A is large)
- More efficient numerical methods exist.

Overdetermined: systems with more equations than unknowns.

$A_{m \times n}$ $(m > n)$

e.g. linear regression: n coefficients to fit m data pts. (x, y)



slope

$y = mx + b$ → y-intercept

2 unknowns

$$y_1 = mx_1 + b$$
$$y_2 = mx_2 + b$$
$$\vdots$$
$$y_m = mx_m + b.$$

m equations

$m > 2 \Rightarrow$ overdetermined

Undetermined : less equations than unknowns ($m < n$)

MATLAB : Solve $Ax = b$

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>> x = A\b
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'invert A

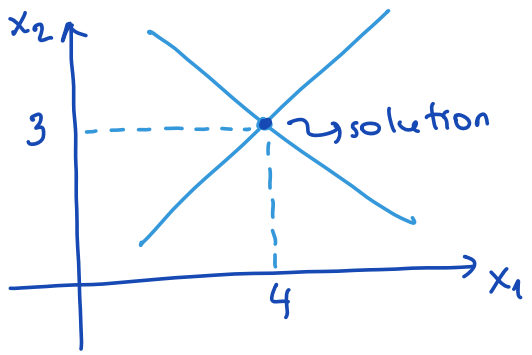
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>> x = inv(A) * b
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Chapter 9 : Gauss Elimination

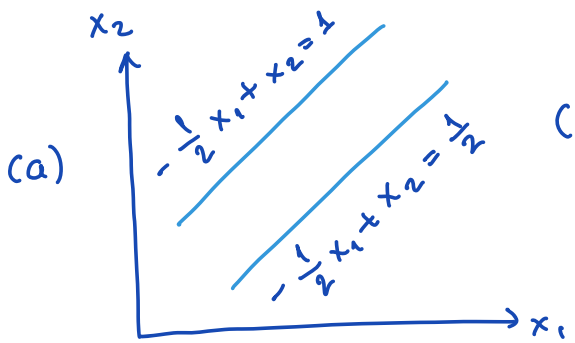
combine equations to eliminate unknowns

EX $3x_1 + 2x_2 = 18$, $-x_1 + 2x_2 = 2$ $\longrightarrow$ $x_2 = -\frac{3}{2}x_1 + 9$, $x_2 = \frac{1}{2}x_1 + 1$ → lines in 2D

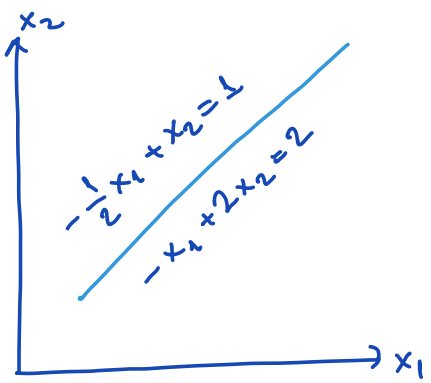
slope, y-intercept



Three Problematic Cases:



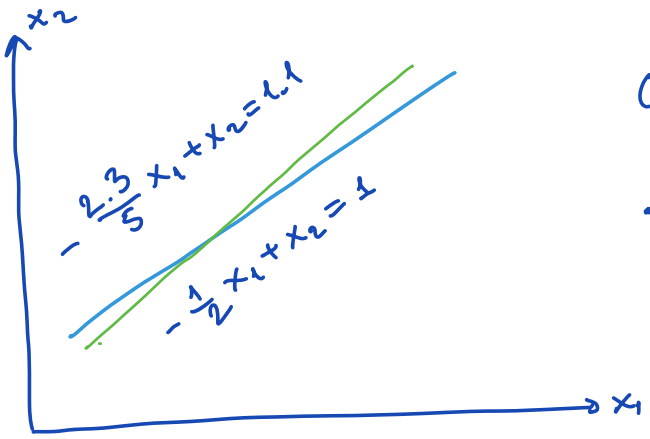
parallel lines
no solution.



coincident lines
infinite solutions

singular cases

(c)



ill-conditioned
(very close to being singular)
- Difficult to identify the exact pt. at which they intersect.

Determinants and Cramer's Rule

- Best suited to small # of eq^{n}s.

Determinants: $Ax = b$

coefficient matrix: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Determinant of this system:

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$(-1)^{i+j}$: $i+j$ → row index, column index

minors

$$D = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

EX Compute the determinant of systems given in (a), (b), (c) (d) above.

(d) $D = \begin{vmatrix} 3 & 2 \\ -1 & 2 \end{vmatrix} = 6+2 = 8$

(a) $D = \begin{vmatrix} -\frac{1}{2} & 1 \\ -\frac{1}{2} & 1 \end{vmatrix} = 0$

(b) $D = \begin{vmatrix} -\frac{1}{2} & 1 \\ -1 & 2 \end{vmatrix} = -\frac{1}{2}(2) - (-1) = 0$

} Singular systems

(c) $D = \begin{vmatrix} -\frac{1}{2} & 1 \\ -\frac{2.3}{5} & 1 \end{vmatrix} = -\frac{1}{2} + \frac{2.3}{5} = -0.04$ ← almost zero.

Cramer's Rule:

3 eq^{n}s, 3 unknowns : $Ax = b$

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}}{D}, \quad D = |A|$$

EX use cramer's rule to solve:

$$0.3x_1 + 0.52x_2 + x_3 = -0.01$$
$$0.5x_1 + x_2 + 1.9x_3 = 0.67$$
$$0.1x_1 + 0.3x_2 + 0.5x_3 = -0.44$$

$$D = 0.3\begin{vmatrix} 1 & 1.9 \\ 0.3 & 0.5 \end{vmatrix} - 0.52\begin{vmatrix} 0.5 & 1.9 \\ 0.1 & 0.5 \end{vmatrix} + \begin{vmatrix} 0.5 & 1 \\ 0.1 & 0.3 \end{vmatrix} = -0.0022$$

$$x_1 = \frac{\begin{vmatrix} -0.01 & 0.52 & 1 \\ 0.67 & 1 & 1.9 \\ -0.44 & 0.3 & 0.5 \end{vmatrix}}{D} = -14.9$$

$$x_2 = \frac{\begin{vmatrix} 0.3 & -0.01 & 1 \\ 0.5 & 0.67 & 1.9 \\ 0.1 & -0.44 & 0.5 \end{vmatrix}}{D} = -29.5$$

$$x_3 = \frac{\begin{vmatrix} 0.3 & 0.52 & -0.01 \\ 0.5 & 1 & 0.67 \\ 0.1 & 0.3 & -0.44 \end{vmatrix}}{D} = 19.8$$

* For more than 3 eq^{n}s, Cramer's rule is impractical since determinants are time consuming.
* More efficient alternatives are used.

Elimination of Unknowns

(1) $a_{11}x_1 + a_{12}x_2 = b_1$

(2) $a_{21}x_1 + a_{22}x_2 = b_2$

$\times a_{21}$ $\quad a_{21}a_{11}x_1 + a_{21}a_{12}x_2 = a_{21}b_1$

$\times a_{11}$ $\quad a_{11}a_{21}x_1 + a_{11}a_{22}x_2 = a_{11}b_2$

$$(a_{21}a_{12} - a_{11}a_{22})x_2 = a_{21}b_1 - a_{11}b_2$$

$$x_2 = \frac{a_{21}b_1 - a_{11}b_2}{a_{21}a_{12} - a_{11}a_{22}} \quad (3)$$

Substitute (3) in (1):

(Tedious for large systems)

$$x_1 = \frac{a_{22} b_1 - a_{12} b_2}{a_{11} a_{22} - a_{21} a_{12}} \qquad (4)$$

using Cramer's rule:

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

(same as (4))

$$x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

(same as (3))

Naive Gauss Elimination

Goal: extend elimination of unknowns method to large sets of eqns by developing a systematic algorithm. to eliminate unknowns & back-substitute.

(Naive: since it does not avoid division by zero.)

General set of n equations:

pivot element a_{11}

$$a_{11} x_1 + a_{12} x_2 + \cdots + a_{1n} x_n = b_1 \quad \leftarrow \text{pivot equation}$$

multiply the 1st eqn by $\frac{a_{21}}{a_{11}}$ and subtract from 2nd eqn.

$$a_{21} x_1 + \cdots + a_{2n} x_n = b_2$$

$$\vdots \qquad\qquad \vdots$$

$$a_{n1} x_1 + \cdots + a_{nn} x_n = b_n$$

division by a_{11} is called as normalization

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

forward elimination

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a''_{33} & b''_3 \end{bmatrix}$$

back substitution:

$$x_3 = \frac{b''_3}{a''_{33}}$$

$$x_2 = \frac{b'_2 - a'_{23} x_3}{a'_{22}}$$

$$x_1 = \frac{b_1 - a_{12} x_2 - a_{13} x_3}{a_{11}}$$

Operation Counting

flops: floating point operations

$$\underbrace{\frac{2n^3}{3} + O(n^2)}_{\text{forward elim.}} + \underbrace{n^2 + O(n)}_{\text{back-subst.}} \xrightarrow{\text{as } n \text{ increases}} \frac{2n^3}{2} + O(n^2)$$

⇒ most of the effort is in forward elimination.

Pivoting:

Given:

$$2x_2 + 3x_3 = 8 \quad \longrightarrow a_{11} = 0$$
$$4x_1 + 6x_2 + 7x_3 = -3$$
$$2x_1 - 3x_2 + 6x_3 = 5$$

Partial pivoting: Determine the coefficient with the largest absolute value in the column below the pivot element. Switch the rows so that the largest element is the pivot.

Complete pivoting: Switch the columns as well.

Ex Use Gauss elimination to solve

$$0.0003x_1 + 3.0000x_2 = 2.0001 \quad (1)$$
$$1.0000x_1 + 1.0000x_2 = 1.0000 \quad (2)$$

Solve with pivot $= 0.0003$ (very close to zero)
Repeat with partial pivoting.

Exact solution: $x_1 = \frac{1}{3}$, $x_2 = \frac{2}{3}$

Sol^n Multiply the first eq^n with $\frac{1}{0.0003}$

$$x_1 + 10000x_2 = 6667$$

subtract from the second eq^n:

$$-9999x_2 = -6666 \Rightarrow x_2 = \frac{2}{3} \rightarrow \text{substitute in eq}^n\ (1)$$

$$x_1 = \frac{2.0001 - 3\left(\frac{2}{3}\right)}{0.0003}$$

very close to each other
Due to subtractive cancellation, the result is sensitive to # significant figures in computation

$$x_1 \simeq 0.3330000$$

Let $\frac{2}{3} = 0.6666667$ (with 7 significant figures)

($x_1 = 0.33333333333334037$ (with 15 significant figures))

(e.g. with $x_2 = 0.667 \Rightarrow x_1 = -3.33$

Solve with partial pivoting:

$$\begin{cases} 1.0000x_1 + 1.0000x_2 = 1.0000 \\ 0.0003x_1 + 3.0000x_2 = 2.0001 \end{cases}$$

multiply 1st eqn 0.0003 and subtract from 2nd eqn:

$$2.9997x_2 = 1.9998$$

$$x_2 = 0.6666\ldots = \frac{2}{3}$$

insert into 1st eqn:

$$\Rightarrow x_1 = \frac{1 - \frac{2}{3}}{1} = \frac{1}{3} \quad \text{(no round-off error)}$$

Determinant Evaluation by Gaussian Elimination

Determinant of a triangular matrix is the product of its diagonal elements

$$D = a_{11}\, a_{22}\, a_{33} \cdots a_{nn}$$

Proof for a 3x3 system:

$$D = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{vmatrix}$$

$$D = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ 0 & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} 0 & a_{23} \\ 0 & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} 0 & a_{22} \\ 0 & 0 \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33}) - 0 + 0$$

* Forward elimination step of Gauss elimination gives an upper triangular system.

* Since the determinant <u>does not change</u> by forward elimination, determinant can be evaluated as:

$$D = a_{11}\, a'_{22}\, a''_{33} \cdots a_{nn}^{(n-1)}$$

$(n-1)$ → # of times elements have been modified.

* Determinant changes sign every time a row is switched.

$$D = a_{11}\, a'_{22}\, a''_{33} \cdots a_{nn}^{(n-1)}\, (-1)^{p}$$

p → # times that rows are switched.